Cryptography

Introduction to Cryptography

Cristhian Iza

School of Mathematical and Computational Sciences
Yachay Tech University

August 19, 2026

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

==========

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Learning Objectives

After this unit you will be able to:

- Explain the difference between cryptology, cryptography and cryptanalysis
- Describe the basic model of symmetric cryptography (Alice, Bob and Oscar)
- Analyse simple historical ciphers (substitution, Caesar, affine)
- Use modular arithmetic and integer rings
- Apply Kerckhoffs' Principle
- Understand recommended key lengths

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Overview

1.1 Overview of Cryptology

Important definitions

- **Cryptology**: the general science that includes cryptography and cryptanalysis
- **Cryptography**: the science of secret writing (hiding the meaning of a message)
- **Cryptanalysis**: the science of **breaking** cryptosystems

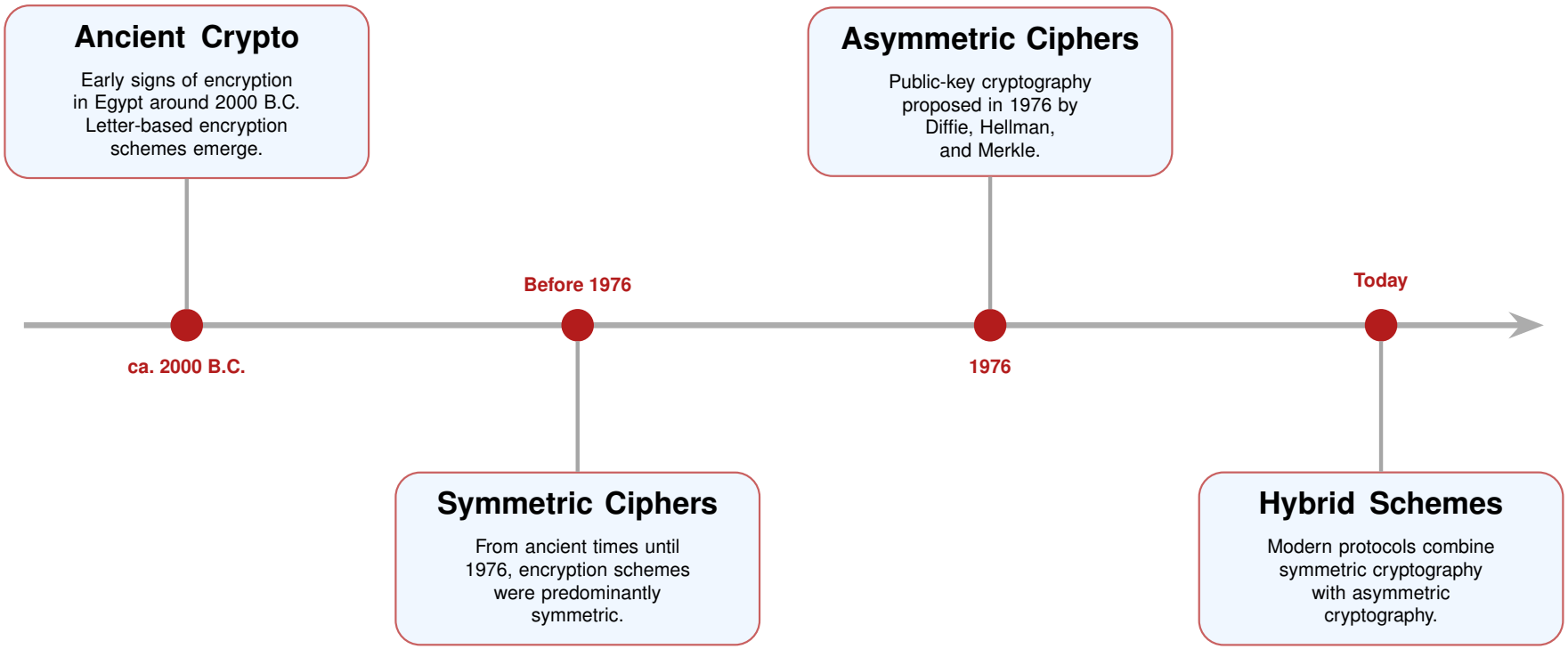
SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Main Branches of Cryptography

- **Symmetric algorithms** (secret-key / single-key)
  - The same key is used for encryption and decryption
  - Used from ancient times until 1976
- **Asymmetric algorithms** (public-key)
  - Introduced by Diffie, Hellman and Merkle in 1976
  - One public key and one private key
- **Cryptographic protocols**
  - They use the algorithms above (example: TLS in web browsers)

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Some Basic Facts



SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Symmetric Cryptography

Communication over an insecure channel

Problem Statement

1. Alice and Bob would like to communicate via an unsecure channel (e.g., WLAN or Internet).
2. A malicious third party Oscar has channel access but should not be able to understand the communication.

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Symmetric Cryptography – Basic Model

- Alice wants to send a message x to Bob
- Oscar can listen to the insecure channel
- Alice encrypts: $y = e_k(x)$
- Bob decrypts: $x = d_k(y)$

Important variables

x	plaintext
y	ciphertext
k	key
K	key space

Important note

The algorithm is public. Only the key must stay secret.

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Symmetric Cryptography

- The key must be transmitted via a **secure channel** between Alice and Bob.
- The secure channel can be realized, e.g., by manually installing the key for WPA or using a human courier.
- The system is only secure if an attacker does not learn K.

Key point

The secure-communication problem is reduced to secure transmission and storage of the key K.

$$d_K(y) = d_K(e_K(x)) = x$$

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Why do we need Cryptanalysis?

- Practical ciphers cannot usually be proven secure.
- Confidence comes from repeated attempts to break them.

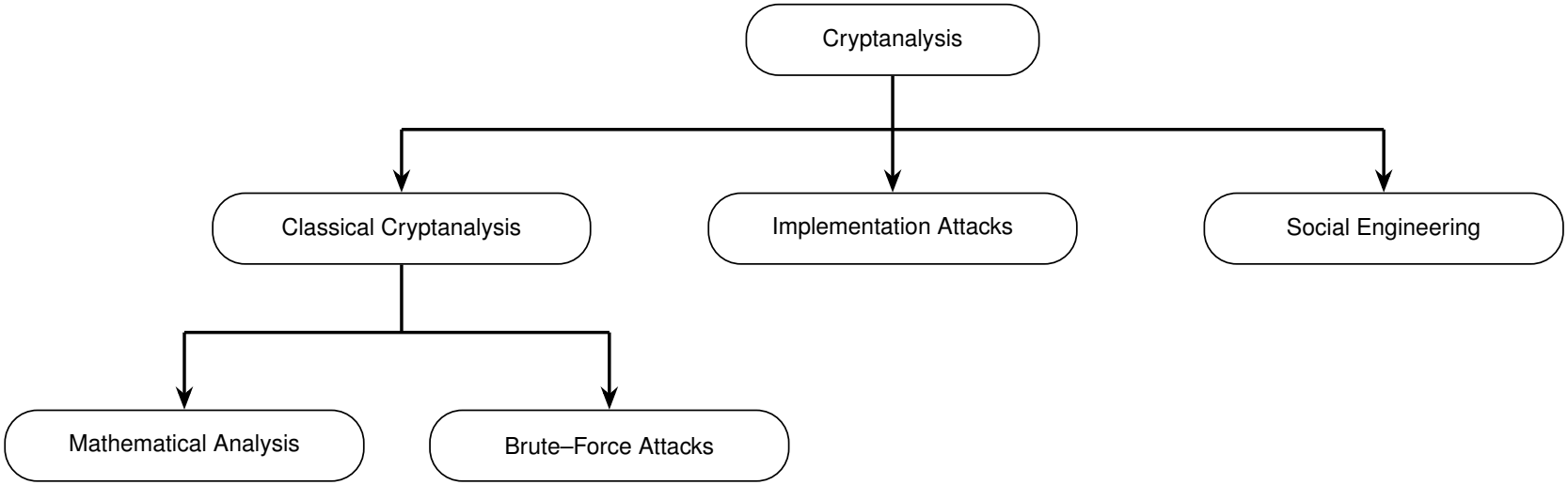
Kerckhoffs' Principle

A cryptosystem should remain secure even if everything is public **except the secret key**.

- Prefer well-known, extensively analyzed ciphers.
- **Security through obscurity is not security.**

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Cryptanalysis: Attacking Cryptosystems



- **Implementation attack:** try to extract the key through reverse engineering or power measurement, e.g., for a banking smart card.
- **Social engineering:** trick a user into giving up a password.

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Brute-Force Attack (Exhaustive Key Search)

- Treats the cipher as a black box.
- Requires one plaintext–ciphertext pair (x_0, y_0).
- Tests all possible keys until:

$$d_K(y_0) \stackrel{?}{=} x_0$$

Key length	Key space	Brute-force resistance
64 bits	2^{64}	Short-term; feasible with sufficient resources.
128 bits	2^{128}	Long-term (several decades without quantum computers).
256 bits	2^{256}	Long-term (designed to resist quantum search better).

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

The Substitution Cipher

The Substitution Cipher

Main idea

Every letter of the alphabet is replaced by another letter according to a substitution table (this table is the key).

Ciphertext example

```
iq ifcc vqqr fb rdq vfllcq na rdq
cfjwhwz hr bnnb hcc
hwwhbsqvqbre hwq vhlq
```

- Key space: $26! \approx 2^{88}$
- This looks strong against brute-force attacks...
- **But it is not secure**

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Attacks on the Substitution Cipher

1. **Brute-force attack (Exhaustive Key Search)**
   - Try all 26! possible tables
   - This is computationally impossible
2. **Letter frequency analysis** (analytical attack)
   - The same plaintext letter always becomes the same ciphertext letter
   - Statistical properties of the language stay in the ciphertext
   - In English: E $\approx$ 13 %, T $\approx$ 9 %, A $\approx$ 8 % ...

Lesson learned

A large key space is **not enough**. A good cipher must hide the statistical properties of the plaintext.

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Cryptanalysis

1.3 Cryptanalysis – How to Break a System

Classical cryptanalysis

- Mathematical / analytical attacks
- Brute-force attacks

Implementation attacks

- Side-channel analysis (power consumption, timing, electromagnetic radiation)

Social engineering

- Bribery, blackmail, tricks
- "Hello, this is the IT department. I need your password. . . "

Basic principle

An attacker always looks for the **weakest point** of the system.

MATHEMATICAL AND COMPUTATIONAL SCIENCES

Kerckhoffs' Principle (1883)

Definition

A cryptosystem must be secure even if the attacker (Oscar) knows **everything** about the system, except the secret key.

Important

- "Security by obscurity" almost always fails
- Historical example: Content Scrambling System (CSS) for DVDs
- Only the key must be kept secret

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

How Many Key Bits Are Enough?

Brute-Force Security

Key length	Key space	Practical security
56–64 bits	2^{56}–2^{64}	Insufficient
112–128 bits	2^{112}–2^{128}	Strong long-term security
256 bits	2^{256}	Very high security margin

- Key length matters when **brute-force search is the best known attack**.
- Increasing computational power gradually reduces the security margin of fixed-size keys.

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Modular Arithmetic

Why Modular Arithmetic?

Core Idea

Modular arithmetic provides the mathematical foundation for many **classical and modern cryptographic systems**.

- **Classical cryptography:** Caesar and affine ciphers naturally operate over integers modulo n.
- **Public-key cryptography:** RSA relies on modular exponentiation and arithmetic in $\mathbb{Z}_n$.
- **Elliptic-curve cryptography:** ECC performs arithmetic over finite fields such as $\mathbb{Z}_p$.

Goal

Cryptography often replaces ordinary arithmetic with **arithmetic over finite sets**.

AND
L SCIENCES

Short Introduction to Modular Arithmetic

- Cryptographic algorithms often operate on sets that are **discrete** and **finite**.
- **A simple example for a finite set in everyday life?**

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Short Introduction to Modular Arithmetic

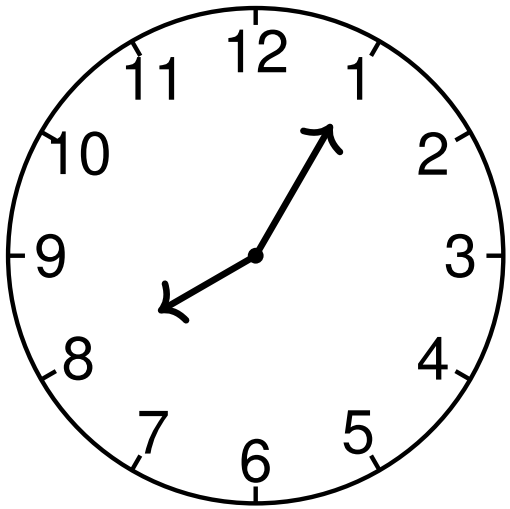
- Cryptographic algorithms often operate on sets that are **discrete** and **finite**.
- A simple example is arithmetic on a **12-hour clock**: after 12, the counting starts again from 1.

Clock Arithmetic

$$12 + 20 = 32$$

but on a 12-hour clock,

$$12 + 20 \equiv ? \pmod{12}$$



SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Introduction to Modular Arithmetic

- Modular arithmetic allows us to compute with a **finite set of integers**.
- After an operation, the result is reduced to a value between 0 and $m - 1$.

Definition of Modulo Operation

For $a, r, m \in \mathbb{Z}$ and $m > 0$,

$$\boxed{a \equiv r \pmod{m}}$$

if m divides $(a - r) \rightarrow [m \mid (a - r)]$

Where:
r is called the remainder
m is called the modulus

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Introduction to Modular Arithmetic

Example

For $a = 13, m = 9, r =?$

$$\boxed{a \equiv r \pmod{m}}$$

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Introduction to Modular Arithmetic

- For any $a \in \mathbb{Z}$ and $m > 0$, it is always possible to write

Computation of the Remainder

$$\boxed{a = q \cdot m + r}$$

, with

$$0 \leq r < m.$$

Since

$$a - r = q \cdot m,$$

m divides $(a - r)$, and therefore

$$\boxed{a \equiv r \pmod{m}}.$$

The remainder satisfies

$$r \in \{0, 1, 2, \ldots, m-1\}.$$

AND
L SCIENCES

Introduction to Modular Arithmetic

- $12 \equiv 3 \pmod 9$, 3 is a valid representative since

$$9 \mid (12 - 3).$$

- $12 \equiv 21 \pmod 9$, 21 is a valid representative since

$$9 \mid (12 - 21).$$

- $12 \equiv -6 \pmod 9$, -6 is a valid representative since

$$9 \mid (12 - (-6)).$$

where the notation $x \mid y$ means "x divides y".
There is a system behind this behavior. The set of numbers

$$\boxed{\{\ldots, -24, -15, -6, 3, 12, 21, 30, \ldots\}}$$

forms what is called an **equivalence class**.

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Properties of Modular Arithmetic (1)

Congruent Representatives

For a fixed modulus m, an integer has infinitely many **congruent representatives**.

Example: mod 9**:**

$$\boxed{12 \equiv 3 \equiv 21 \equiv -6 \pmod 9}$$

because

$$12 - 3 = 9, \qquad 12 - 21 = -9, \qquad 12 - (-6) = 18.$$

All these differences are multiples of 9.

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Equivalence Classes Modulo 5

The congruence classes modulo 5 are:

$$\mathbb{Z}_5 = \{[0]_5, [1]_5, [2]_5, [3]_5, [4]_5\}$$

where:

$$\begin{aligned}
[0]_5 &= \{\ldots, -10, -5, 0, 5, 10, 15, \ldots\}, \\
[1]_5 &= \{\ldots, -9, -4, 1, 6, 11, 16, \ldots\}, \\
[2]_5 &= \{\ldots, -8, -3, 2, 7, 12, 17, \ldots\}, \\
[3]_5 &= \{\ldots, -7, -2, 3, 8, 13, 18, \ldots\}, \\
[4]_5 &= \{\ldots, -6, -1, 4, 9, 14, 19, \ldots\}.
\end{aligned}$$

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Properties of Modular Arithmetic (2)

Which remainder do we choose?

- By convention, we usually choose the unique representative r satisfying

$$a = qm + r, \qquad 0 \leq r \leq m - 1.$$

Example:

$$12 = 1 \cdot 9 + 3 \quad \Rightarrow \quad r = 3.$$

Remark

This is a convention. Algorithmically, one may use any valid congruent representative while computing cryptographic functions.

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Properties of Modular Arithmetic (2)

Which Remainder Do We Choose?

The use the value r such that

$$\boxed{0 \leq r < m}.$$

This value is called the **least non-negative remainder.**

Example: $a = 12$, $m = 9$

$$12 = 1 \cdot 9 + 3,$$

therefore,

$$\boxed{12 \bmod 9 = 3 \bmod 9}$$

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Efficient Modular Computation

Computational Principle

Reduce modulo m as early as possible.
The final result is unchanged, but the computation is simpler.

Example: Compute $3^8 \bmod 7$

Direct method

$$3^8 = 6561$$

$$6561 \equiv 2 \pmod 7$$

$$\boxed{3^8 \equiv 2 \pmod 7}$$

Large intermediate value: 6561.

Modular reduction

$$3^8 = 3^4 \cdot 3^4$$

$$81 \equiv 4 \pmod 7$$

$$3^8 \equiv 4 \cdot 4 = 16 \equiv 2 \pmod 7$$

Intermediate values remain small.

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Efficient Modular Exponentiation

Challenge

Avoid computing the complete power. Use repeated squaring and modular reduction.

Compute:

1. $2^{20} \bmod 7$
2. $5^{16} \bmod 11$
3. $7^{13} \bmod 12$

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Properties of Modular Arithmetic (3)

Modular Division

Division modulo m is performed by multiplying by a **multiplicative inverse**:

$$\boxed{\frac{b}{a} \equiv b\,a^{-1} \pmod{m}}.$$

An inverse a^{-1} satisfies

$$\boxed{a\,a^{-1} \equiv 1 \pmod{m}}.$$

Important:

$$\boxed{a^{-1} \text{ exists modulo } m \iff \gcd(a, m) = 1.}$$

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Practice: Does the Modular Inverse Exist?

Existence Condition

An element a has a multiplicative inverse modulo m if and only if

$$\boxed{\gcd(a, m) = 1}$$

Determine whether the inverse exists:

1. $3^{-1} \pmod 7$
2. $6^{-1} \pmod 9$
3. $5^{-1} \pmod{12}$
4. $8^{-1} \pmod{15}$

First Step

Always compute $\gcd(a, m)$ before searching for the inverse.

AND
L SCIENCES

Computing Modular Inverses

Definition

The multiplicative inverse of a modulo m is a value a^{-1} such that

$$\boxed{a\,a^{-1} \equiv 1 \pmod{m}}$$

provided that $\gcd(a, m) = 1$.

Compute the inverse

1. Find $3^{-1} \pmod{7}$

Strategy

Find x such that

$$a\,x \equiv 1 \pmod{m}.$$

AND
L SCIENCES

Computing Modular Inverses

Condition

The inverse of a modulo m exists only if

$$\boxed{\gcd(a, m) = 1}$$

Example:

$$\gcd(7, 26) = 1$$

Therefore,

$$\boxed{7^{-1} \pmod{26} \text{ exists}}$$

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Finding s and t

If

$$\gcd(a, m) = 1,$$

find integers s and t such that

$$\boxed{as + mt = 1}$$

For $a = 7$ and $m = 26$:

$$1 = 7(-11) + 26(3)$$

Therefore,

$$\boxed{s = -11}, \qquad \boxed{t = 3}$$

Key Point

s multiplies a; t multiplies m.

Multiplicative Inverse

From

$$1 = 7(-11) + 26(3),$$

the coefficient of 7 is

$$\boxed{s = -11}.$$

Therefore,

$$7^{-1} \equiv -11 \pmod{26}.$$

Since

$$-11 \equiv 15 \pmod{26},$$

$$\boxed{7^{-1} \equiv 15 \pmod{26}}$$

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Finding the Coefficients s and t

Goal: Find s and t such that

$$\boxed{7s + 26t = 1}$$

1. Euclidean Algorithm

$$26 = 3(7) + 5 \rightarrow 5 = 26 - 3(7)$$

$$7 = 1(5) + 2 \rightarrow 2 = 7 - 1(5)$$

$$5 = 2(2) + 1 \rightarrow 1 = 5 - 2(2)$$

2. Back-substitution

$$1 = 5 - 2(2)$$

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Finding the Coefficients s and t (cont.)

$$1 = 5 - 2(7 - 5)$$

$$1 = 3(5) - 2(7)$$

$$1 = 3(26 - 3(7)) - 2(7)$$

$$\boxed{1 = 3(26) - 11(7)}$$

3. Identify the coefficients

$$1 = 7(-11) + 26(3)$$

$$\boxed{s = -11} \qquad \boxed{t = 3}$$

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Properties of Modular Arithmetic (3)

Example: What is $5/7 \pmod 9$?

Integer Rings $\mathbb{Z}_m$

Definition

The integer ring $\mathbb{Z}_m$ contains the integers

$$\boxed{\mathbb{Z}_m = \{0, 1, 2, \ldots, m-1\}}$$

with addition and multiplication performed modulo m.

For $a, b \in \mathbb{Z}_m$:

$$a + b \equiv c \pmod{m}, \qquad c \in \mathbb{Z}_m$$

$$a \cdot b \equiv d \pmod{m}, \qquad d \in \mathbb{Z}_m$$

Key Idea

Operations remain inside the finite set $\mathbb{Z}_m$.

AND
L SCIENCES

Example: The Ring $\mathbb{Z}_9$

For $m = 9$,

$$\boxed{\mathbb{Z}_9 = \{0, 1, 2, 3, 4, 5, 6, 7, 8\}}$$

Addition:

$$6 + 8 = 14$$

Reduce modulo 9:

$$14 \equiv \boxed{5} \pmod 9$$

Multiplication:

$$6 \cdot 8 = 48$$

$$48 \equiv \boxed{3} \pmod 9$$

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Properties of $\mathbb{Z}_m$

For $a, b, c \in \mathbb{Z}_m$:

- **Closure:** $a + b$ and $a \cdot b$ remain in $\mathbb{Z}_m$.
- **Associativity:**

$$a + (b + c) = (a + b) + c$$

$$a(bc) = (ab)c$$

- **Additive identity:**

$$a + 0 \equiv a \pmod{m}$$

- **Multiplicative identity:**

$$a \cdot 1 \equiv a \pmod{m}$$

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Inverses in $\mathbb{Z}_m$

Additive Inverse

Every $a \in \mathbb{Z}_m$ has an additive inverse $-a$:

$$\boxed{a + (-a) \equiv 0 \pmod{m}}$$

Multiplicative Inverse

A multiplicative inverse a^{-1} satisfies

$$\boxed{a \cdot a^{-1} \equiv 1 \pmod{m}}$$

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Inverses in $\mathbb{Z}_m$ (cont.)

Important Difference

The additive inverse always exists.

The multiplicative inverse exists only for some elements.

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

When Does the Multiplicative Inverse Exist?

Condition

An element a has a multiplicative inverse modulo m if and only if

$$\boxed{\gcd(a, m) = 1}$$

Example 1:

$$\gcd(15, 26) = 1$$

Therefore,

$$\boxed{15^{-1} \pmod{26} \text{ exists}}$$

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

When Does the Multiplicative Inverse Exist? (cont.)

Example 2:

$$\gcd(14, 26) = 2 \neq 1$$

Therefore,

$$\boxed{14^{-1} \pmod{26} \text{ does not exist}}$$

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Integer Rings

- $\mathbb{Z}_m$ contains
$$\{0, 1, \ldots, m-1\}.$$
- Addition and multiplication are performed modulo m.
- Additive inverses always exist.
- Multiplicative inverses exist only when
$$\boxed{\gcd(a, m) = 1}.$$

Cryptographic Relevance

Arithmetic in $\mathbb{Z}_m$ is a fundamental tool in public-key cryptography.

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Integer Rings $\mathbb{Z}_m$

Definition

The integer ring $\mathbb{Z}_m$ consists of:

1. The set $\{0, 1, 2, \ldots, m-1\}$
2. Two operations $+$ and $\times$ modulo m

Important properties:

- Closed under $+$ and $\times$
- Additive identity: 0
- Multiplicative identity: 1
- Additive inverse always exists
- Multiplicative inverse exists **if and only if** $\gcd(a, m) = 1$

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Shift Cipher (Caesar Cipher)

Definition

Let $x, y, k \in \mathbb{Z}_{26}$.

$$\text{Encryption:} \quad e_k(x) \equiv x + k \pmod{26}$$
$$\text{Decryption:} \quad d_k(y) \equiv y - k \pmod{26}$$

Example: $k = 17$, plaintext "ATTACK"

$$0, 19, 19, 0, 2, 10 \quad \rightarrow \quad 17, 10, 10, 17, 19, 1 \quad = \text{RKKRTB}$$

Security

Only 26 possible keys $\Rightarrow$ very easy to break with brute-force or frequency analysis.

AND COMPUTATIONAL SCIENCES

Affine Cipher

Definition

Let $x, y, a, b \in \mathbb{Z}_{26}$ and $\gcd(a, 26) = 1$.

$$\text{Encryption:} \quad e_k(x) \equiv a \cdot x + b \pmod{26}$$

$$\text{Decryption:} \quad d_k(y) \equiv a^{-1}(y - b) \pmod{26}$$

Key: $k = (a, b)$

Key space: $12 \times 26 = 312$ (still very small)

Conclusion

Both the Caesar cipher and the affine cipher are insecure. They are only useful for teaching.

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Lessons Learned

- Never design your own crypto algorithm without experienced cryptanalysts
- Do not use unproven algorithms or protocols
- An attacker always looks for the weakest point
- A large key space alone does not guarantee security
- Recommended symmetric key lengths:
  - 64 bits $\rightarrow$ insecure
  - 112–128 bits $\rightarrow$ long-term security
  - 256 bits $\rightarrow$ resistant to known quantum algorithms
- Modular arithmetic is the base of almost all modern cryptography

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Summary of Unit 1

1. Cryptology = Cryptography + Cryptanalysis
2. Symmetric model: Alice, Bob, Oscar + secret key
3. Historical ciphers (substitution, Caesar, affine) are only for teaching
4. Kerckhoffs' Principle: the algorithm is public
5. Modular arithmetic and rings $\mathbb{Z}_m$ are fundamental tools
6. Real security starts with modern algorithms (next units)

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Problems

Problem 1: Shift Cipher Attack

We received the following ciphertext, which was encoded with a **shift cipher**:

```
JS RFHTSIT QF RJRTWNF LZFWIF XJHWJYTX VZJ JQ
YNJRUT ST UZJIJ GTWWFW
```

1. Perform an attack against the cipher using a **letter frequency count**.
2. Determine the shift value used as the key.
3. Recover the plaintext.
4. Explain briefly how frequency analysis helped you identify the key.

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Problem 2: Modular Arithmetic

Modular arithmetic is the basis of many cryptographic systems.

Compute the following without a calculator:

1. $$17 \cdot 23 \bmod 11$$
2. $$4 \cdot 31 \bmod 13$$
3. $$7 + 19 \bmod 9$$
4. $$-13 \cdot 5 \bmod 17$$

Requirement

Reduce intermediate values whenever possible.

Problems 3: Modular Division

Compute without a calculator:

1. $$\frac{1}{4} \pmod{11}$$
2. $$\frac{1}{6} \pmod{13}$$
3. $$3 \cdot \frac{2}{5} \pmod{11}$$

Before Computing

For every denominator a, first verify

$$\boxed{\gcd(a, m) = 1}.$$

Problem 4: Integer Rings

Consider the ring

$$\mathbb{Z}_5 = \{0, 1, 2, 3, 4\}.$$

1. Construct the **addition table** for $\mathbb{Z}_5$.
2. Construct the **multiplication table** for $\mathbb{Z}_5$.
3. Identify the elements that have a **multiplicative inverse**.
4. For every nonzero element a, compute
$$\gcd(a, 5).$$
5. Explain why every nonzero element of $\mathbb{Z}_5$ has a multiplicative inverse.

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Problem 5: Multiplicative Inverses in Rings

Consider the rings

$$\mathbb{Z}_6 \qquad \text{and} \qquad \mathbb{Z}_7.$$

1. Identify all nonzero elements in $\mathbb{Z}_6$ that do **not** have a multiplicative inverse.
2. For each case, compute
$$\gcd(a, 6).$$
3. Determine whether every nonzero element in $\mathbb{Z}_7$ has an inverse.
4. Explain your answer using
$$\boxed{\gcd(a, m) = 1}.$$

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Problem 6: Modular Exponentiation

Compute x as far as possible **without a calculator**.
Where appropriate, reduce intermediate results modulo 13.

1. $$x = 4^2 \bmod 13$$
2. $$x = 4^5 \bmod 13$$
3. $$x = 4^{10} \bmod 13$$
4. $$x = 5^{20} \bmod 13$$
5. Solve
$$4^x \equiv 3 \pmod{13}.$$

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES

Cryptography

Introduction to Cryptography

Cristhian Iza

School of Mathematical and Computational Sciences
Yachay Tech University

August 19, 2026

SCHOOL OF MATHEMATICAL AND COMPUTATIONAL SCIENCES